

Security attacks can be passive or active

If it doesn't noticeably affect the user it is passive(Ex: Unauthorized reading of a message)

Otherwise, it is active(Ex: Deletion of a file)

Crypto algo areas:

- Symmetric encryption: hide any data
- Unsymmetric encryption: hide small data
- Data integrity algos: protect data from modification
- Authentication protocols: verify identity

Computer security aims to preserve Confidentiality, Integrity and Availability(CIA) of data and system and arguably even ensure Authenticity and Accountability